NGUYEN ANH MINH

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Summary

Software Engineer with hands-on experience in **Golang, Node.js, React.js, and DevOps**, including gRPC integration. Proficient in **Agile/Scrum, Git, RESTful APIs, Docker, Kubernetes, CI/CD pipelines**, and cloud platforms such as **AWS and GCP**. A proactive team player with strong experience in code reviews, unit testing, and system design.

Skills

Programming Languages: Golang, TypeScript, Nodejs, Java, Python
Frameworks: Google frameworks, go-kit, Kratos, NestJS, ASP.NET, Reactjs, Next.js, Angular,Express
Databases : SQL Server, PostgreSQL, Redis, MongoDB
DevOps / Infrastructure : Jenkins, Grafana, Prometheus, Jaeger, Envoy, NGINX, Kong, ArgoCD, Docker, Kubernetes, Helm, Terraform
Clouds: AWS, GCP
Others: OOP, Agile/Scrum, Microservices, CQRS, DDD, Outbox Pattern, Saga Pattern, Event Bus, Clean Architect, Hexagon Pattern, Factory Pattern, Singleton Patton ,...

Experience

Software Engineering **01/2025 – Present**
Viet My School — Ho Chi Minh, Vietnam

- Architected and led the development of a microservices-based school management platform **serving ~5,000 concurrent users (~2,000 RPS peak)**, leveraging gRPC for low-latency internal communication and **Kafka for event-driven workflows**; configured **Envoy Proxy for gRPC–HTTP** transcoding and centralized traffic management.
- Designed and implemented a scalable search and reporting system using **Elasticsearch**, integrating **CDC via the Outbox Pattern** to ensure reliable data synchronization from relational databases; handled high-volume queries (student records, payment history, activity logs) with optimized indexing and query tuning, **reducing P95 search latency from 450ms to 180ms**.
- Introduced a **multi-layer caching strategy with Redis** (cache-aside pattern) to reduce database pressure under peak load; implemented cache invalidation policies and mutex-based protection against cache stampede, **reducing P95 API latency by 40% and lowering DB load by ~35%**.
- Deployed and operated the system on Kubernetes, **authoring production-ready Helm charts with autoscaling (HPA)**, resource tuning, and environment segregation; **achieved 99% uptime in staging load tests (k6-based)** and maintained stable performance during peak concurrency scenarios.
- Built and maintained a **GitOps-driven CI/CD pipeline using GitHub Actions**, **Docker Hub**, and **ArgoCD**, enabling automated build–test–deploy workflows with rollback capability; **reduced deployment lead time by 60%** and improved release .

Software Engineering **1/2024- 10/2024**
Freelancer | E-Commerce Project — Ho Chi Minh, Vietnam

- Designed asynchronous background workflows to **crawl over 1k records** (orders, products, catalogs) daily from Shopee, Lazada, and TikTok with **95% success rate**.
- Implemented the **Outbox Pattern** to handle webhooks from multiple platforms and ensured **distributed locks** and consistency between services.
- Implemented distributed job workflows with distributed locking, **ensuring single-run execution (100% accuracy)**, cutting job conflicts.
- Designed combo box and improved **Redis cache** query of shared combo box to **reduce latency by 40%** and **reduce workload by 40%** for simple data.
- **Performed functions by 60%** with performing suitable **queries** and designed solution for multiple notification with high architecture.

Backend Developer Intern
FPT Software | Online Course Marketplace Platform — Ho Chi Minh, Vietnam **9/2023 – 12/2023**

- Refactored **video download API** by offloading heavy processing (file download & compression) to **background workers** using **BullMQ**, preventing main thread blocking and improving API response time under concurrent requests.
- Separated **CPU-bound tasks** (video processing) from I/O-bound request handling by leveraging **Worker Threads**, reducing event loop lag during peak traffic.
- Implemented queue-based job processing for long-running tasks instead of handling them synchronously in the request lifecycle, **stabilizing API latency and minimizing timeout risks.**
- Monitored event loop delay and optimized **async flow** (avoiding sync fs operations, large JSON parsing), improving responsiveness under burst traffic.
- Applied **basic concurrency control** (rate limiting & Redis-based locking) on high-frequency endpoints to prevent resource exhaustion during spikes.
- Reduced memory pressure by cleaning up unused object references in long-running **worker processes** and limiting large in-memory buffers during file handling.

Educations

FPT University

Ho Chi Minh, Viet Nam

Bachelor's degree | Software Engineering | 3.1/4.0 GPA | Link

01/2022 – 12/2025

Soft Skills

- English : Good Comunication , Read and Write docs ~ ielts 6.0
- Good teamwwork and freindly